

1 Study Design

1.1 Usability Goals

1.1.1 Understandability of Voice Communication Settings

This goal evaluates whether users can understand and correctly use Discord's voice settings, specifically push-to-talk and voice activity, to control when they are heard. Discord operates across many different use contexts, including noisy environments, so users need to be able to identify the right settings, apply them without making errors, and recover if something goes wrong. This goal is relevant to both new and experienced users, as incorrect voice settings can cause unintended broadcasting during group calls.

1.1.2 Efficiency and Findability of Communication Hierarchy

This goal examines how quickly and easily users can complete basic messaging tasks and whether they understand the structural difference between servers and channels in Discord. A server acts as a container for multiple channels, and confusing the two could be an issue for users. The study includes tasks that require navigating between specific channels rather than just any available space in a server.

1.1.3 Findability of Server Management Features

This goal determines whether users can locate and complete server management tasks within Discord, including creating a server and managing identity settings that apply only within a specific server. Discord separates global username settings from server-specific display names, which is not obvious at first glance. This goal focuses on whether users can find and understand these distinct subsystems without guidance.

Goals 1.1.2 and 1.1.3 both examine findability within Discord's interface structure, but they focus on different areas: Goal 1.1.2 focuses on finding and using communication spaces such as servers, channels, direct messages, and search, while Goal 1.1.3 focuses on finding server management and identity settings.

1.2 Methodology

All three concepts are measured using the same two methods: direct observation with screen recording and the think-aloud protocol. This approach allows both quantitative metrics (task completion, time, errors) and qualitative data (verbal comments, signs of confusion) to be captured for each concept. Across all concepts, a mistake was defined as an observable action that moved the participant away from the correct solution or showed a misunderstanding of the interface structure. If the participant makes a mistake but later completes the task correctly, the task is still marked as completed, but the mistake is recorded separately.

1.2.1 Concept 1: Understandability of Voice Communication Settings

This concept focuses on how well users understand Discord's voice settings and whether they can prevent unintended broadcasting. The study measures whether users recognise the difference between push-to-talk and voice activity, and whether they can apply the correct setting to match their situation.

Quantitative data:

- Whether the step was completed successfully: 1 = completed, 0 = not completed.
- Time taken to complete the task
- Number of observable mistakes made during the task
- Whether assistance or a clue was required
- Whether the participant gave up: 1 = gave up, 0 = did not give up.

Qualitative data:

- Verbal comments and questions made during the task (think-aloud)
- Observed signs of confusion or frustration

Specific mistake examples :

- pressing the microphone mute button instead of changing transmission mode;
- adjusting the microphone input volume instead of changing the transmission setting;
- changing speaker/output settings instead of microphone transmission settings;
- turning on unrelated options, such as voice isolation, stream attenuation, deafen, or sound settings, when these do not solve the task;
- selecting the wrong mode, for example, using voice activity when the task requires controlled broadcasting;
- finding push-to-talk but not understanding that a key must be pressed to broadcast;
- Repeatedly checking the same setting without making progress;
- requiring researcher assistance or a clue to locate or understand the relevant setting.

1.2.2 Concept 2: Efficiency and Findability of Communication Hierarchy

This concept examines whether participants can accurately navigate Discord's communication structure across servers, channels, direct messages, mentions, and message search. The aim is not only to measure whether users can complete basic

messaging actions, but also whether they understand where those actions take place within Discord's hierarchy. The main usability issue tested is the separation of communication across different interface areas. For example, a participant may find the correct server but still confuse it with a channel, or they may know how to send a direct message but struggle to locate a previous message or shared link. Each sub-task was observed separately to identify whether difficulties were caused by general navigation, search, direct messaging, mentions, or the specific server/channel hierarchy problem.

Quantitative data:

- Whether the step was completed successfully: 1 = completed, 0 = not completed.
- Whether the participant gave up: 1 = gave up, 0 = did not give up.
- Time taken to complete the step.
- Number of observable mistakes made during the step.
- Whether researcher assistance or a clue was required.

Timing will be recorded as raw observed time first. The time will not be interpreted on its own because a short time does not automatically mean success. A participant may complete a task quickly but post the message in the wrong place. Task time will be analysed together with completion status, mistakes, and observations.

Mistakes counted for this concept:

- Wrong location – the participant sends a message in the wrong server, channel, direct message, or conversation.
- Server/channel confusion – the participant treats a server and a channel as the same location.
- Wrong search or navigation route – the participant uses a search field, menu, profile, or private message route that does not match the task goal and moves them away from the required location.
- Random clicking – the participant clicks several unrelated icons or menus without a clear strategy.
- Assistance required – the participant needs a researcher's clue to continue or correct their route.

Qualitative data:

- Verbal comments and questions made during the task (think-aloud)
- Observed confusion, for example, the participant is guessing rather than understanding the interface.

1.2.3 Concept 3: Findability of Server Management Features

This concept tests whether users can find and use Discord's server-creation and identity-management features. The focus is on two separate subsystems: the server creation flow and the server-specific nickname setting. These are located in different parts of the interface, and it is not immediately obvious that changing a display name inside one server does not affect the username shown elsewhere.

The task is broken down into two observable steps:

Step 1: Create a new server from scratch and name it "Work Project".

Step 2: Change the display name inside that specific server to "Participant N" (their participant number) and check that the original username remains unchanged in all other servers.

Each step is tracked separately. If a participant completes Step 1 but cannot find the nickname option in Step 2, this points to a specific findability issue within the server settings subsystem. If they complete Step 2 but are unsure whether it affected other servers, this shows a gap in understanding how identity settings work across Discord.

Quantitative data:

- Whether the step was completed successfully: 1 = completed, 0 = not completed.
- Whether the participant gave up: 1 = gave up, 0 = did not give up.
- Time taken to locate the server creation option and the server-specific nickname setting
- Number of observable mistakes made during the step.

Qualitative data:

- Verbal comments and questions during the task (think-aloud)
- Observed signs of confusion, for example, navigating to account settings instead of server settings, or giving up on a step

Mistakes counted for this concept:

- opening server settings when the task requires changing the user's server-specific display name
- editing the server name instead of the user's display name
- editing the global Discord display name instead of the server-specific nickname
- Looking in roles, members, invites, access, or account settings without finding the correct option
- changing the name but failing to verify whether the change applies only to the selected server
- requiring researcher assistance to locate the correct profile or nickname setting.

1.3 User Profiles

Since the three goals focus on features that require some prior familiarity with digital communication platforms, only experienced Discord users were selected. Novice users were excluded because the study aimed to test whether interface problems still occur among users who already understand the basic purpose of Discord. This makes the findings more useful for identifying design issues in specific features rather than general onboarding problems.

Profile: Experienced User

Characteristics:

- Has used Discord regularly for at least 5 hours per week
- Uses the platform weekly or daily for communication
- Has experience joining multiple servers
- Has some experience exploring or adjusting server settings

Pre-study screening questions:

The following questions were asked before the session to confirm that the participant fits the profile:

1. Have you ever used Discord?
 - Yes / No
2. If yes, approximately how many hours per week do you use it?
 - Under 5 / Over 5
3. What do you primarily use Discord for?
 - Gaming / Work / Study / Hobby / Other:

1.4 Procedure

1.4.1 Ethics

Before taking part, all external participants were given an information sheet explaining the purpose of the study, the tasks involved, and their rights. Informed consent was obtained via a consent form that outlined how data would be stored and used. For participants who took part in the class exercise, the same information and ethical principles applied, but a formal signed consent form was not required, in line with course guidelines. No personal data was collected, and all results were anonymised.

Preparation

Before any session began, the following was set up in advance to keep conditions consistent across all participants:

- Two Discord servers were created: one named "Usability Testing" containing a general channel and a channel also named "Usability Testing", and a second server used as context for the identity verification step.
- A test account (ut_test1) was prepared and pre-logged in on the device used during the study.
- The test account was already a member of the relevant servers
- Screen recording software was set up before each session.
- A note-taking sheet was prepared to log errors, timestamps, and observations per task step.

Each participant completed all three tasks in the same order. This was intentional: Task 1 introduces them to Discord's voice subsystem, Task 2 tests navigation within the messaging subsystem, and Task 3 moves to the server management subsystem. Because each task covers a different part of the interface, completing an earlier task does not give the participant an advantage in the next one. However, it does mean all participants enter each task at the same point, which limits the bias that would come from participants who explored the app freely beforehand.

1.4.2 Task 1: Voice Communication Settings

The researcher and one participant are present for this task. It takes approximately 5 to 10 minutes.

The participant is given the following scenario:

"You are in a voice call with your coursework group mates, but you are in a very noisy environment. Your goal is to make sure you do not broadcast background noise unintentionally."

The researcher notes:

- Whether the participant navigates directly to voice settings or searches elsewhere first
- Which mode they select first, and whether they change it
- Any errors, such as adjusting the input volume instead of the transmission mode
- Time from opening settings to completing the adjustment

1.4.3 Task 2: Server/Channel Hierarchy and Findability

The participant is asked to complete five communication-related sub-tasks in Discord. The setup uses two servers ("Usability testing" and "UOP Modules"), one of which ("UOP Modules") contains a channel with the same name as the other server. This is intentional, as it creates a real navigational challenge rather than a straightforward search.

The participant completes the following steps:

Step 2a: Find the server named "Usability Testing" and send the message "hello" in the general text channel.

Step 2b: Find the Zoom link sent by 🎓 Shelley Usher on 28/01/2026.

Step 2c: From the user account ut_test1, send a direct message to reni4621.

Step 2d: Mention reni4621 in the Usability Testing server.

Step 2e: Find the channel named "Usability Testing" and send a message inside that channel.

The researcher tracks each sub-task separately and records:

- whether the participant completed the step successfully
- the actual time taken for each step
- the number and type of mistakes
- whether the participant needed help or a clue
- whether the participant confused a server, channel, direct message, or user profile
- verbal comments and signs of hesitation during the think-aloud process.

The participant is asked to think aloud throughout the task, and the session is screen-recorded. If a participant sends a message in the wrong place, this is recorded as a mistake, and the task is marked as incomplete unless the participant independently recognises the error and corrects it within the task attempt.

1.4.4 Task 3: Server Management and Identity

The researcher and one participant are present for this task. It takes approximately 5 to 10 minutes.

The participant is given the following scenario:

"You are starting a new work project and want to create a dedicated space for your group."

They are then asked to complete the following steps:

Step 1: Create a new server from scratch and name it "Work Project".

Step 2: Change the display name inside this specific server to "Participant N" (their participant number) and check that the original username remains unchanged in all other servers.

Each step is observed individually. The researcher tracks:

- Whether the participant finds the server creation button without help
- Whether they look in account settings instead of server settings when attempting to change the display name
- Whether they understand that the change only applies to one server
- Time taken per step and any errors made

All participants attempted both steps regardless of errors. If a participant gives up at a specific step, the researcher notes the exact point and reason. This is useful for identifying where the interface fails, as a pattern of participants giving up at the same step points to a clear usability issue.

The session is screen-recorded, and the participant is asked to think aloud throughout.

2. Results and Implications

2.1 Results

Before analysing the results by concept, two types of data were analysed: quantitative task-performance data and qualitative think-aloud/observation data.

The quantitative data were analysed using descriptive statistics only. For each task or sub-task, the analysis considered completion rate, time on task, benchmark comparison, number of mistakes, whether assistance was required, and whether the participant gave up. The qualitative data were analysed using a simple coding approach. Screen recordings, researcher notes, and think-aloud comments were reviewed for repeated behaviours or statements. The codes were then compared with the quantitative results to explain why participants succeeded, failed, took longer than expected, or needed assistance.

"Gave up" was recorded as a binary quantitative metric the surrounding behaviour was also examined qualitatively. The binary value shows whether abandonment occurred, while the qualitative notes explain why it occurred and what interface problems appeared before the participant stopped.

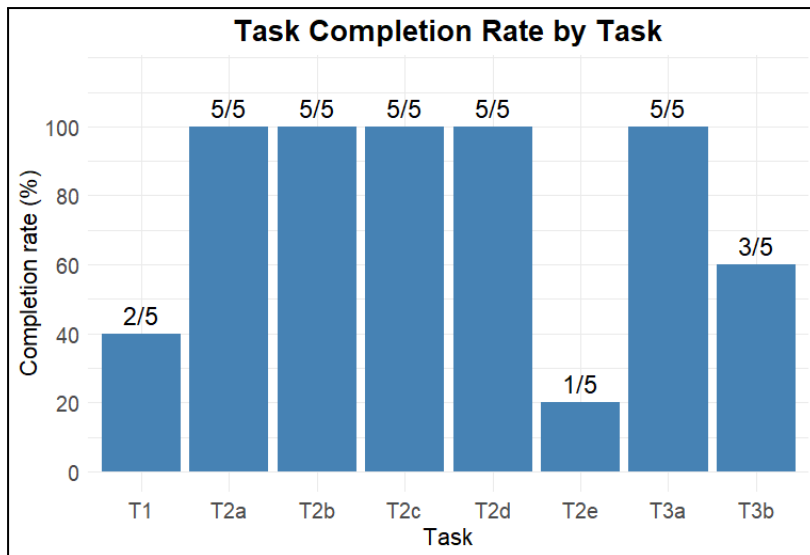


Figure 2.1: Overall completion rate across all tasks

Figure 2.1 provides an overview of task performance across the full study. It shows that participants performed well on familiar communication actions, especially T2a–T2d and T3a, but struggled with tasks that required understanding less visible system distinctions. The weakest results were found in T1, T2e, and T3b. These tasks required participants to distinguish between microphone control and voice transmission mode, between a server and a channel, and between global identity settings and server-specific identity settings.

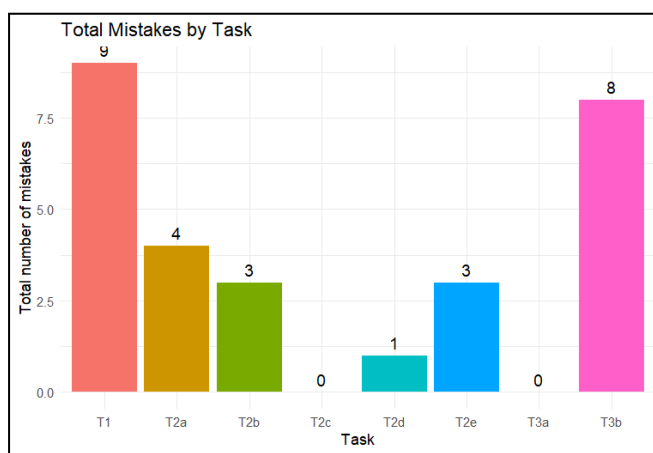


Figure 2.1.2: Overall total mistakes by task

2.1.1 Concept 1: Understandability of Voice Communication Settings

Concept 1 was assessed through Task 1, which required participants to configure Discord's voice settings to avoid broadcasting background noise unintentionally. The analysis focused on whether participants could distinguish between microphone control and voice transmission mode, particularly push-to-talk and voice activity.

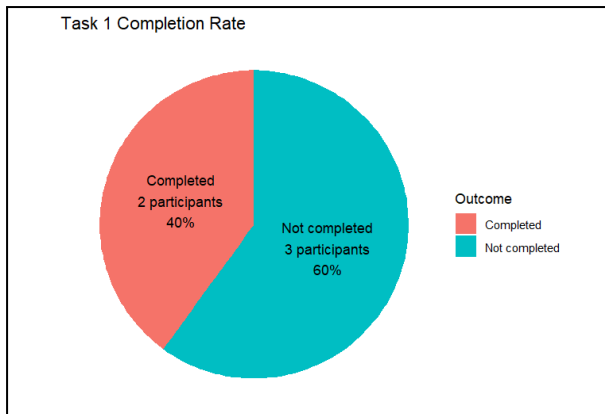


Figure 2.1.1. Task 1 completion rate

Figure 2.1.1 shows that only two of the five participants completed Task 1, while three gave up or failed to achieve a working configuration. This low completion rate suggests that understanding the voice settings was difficult for users.

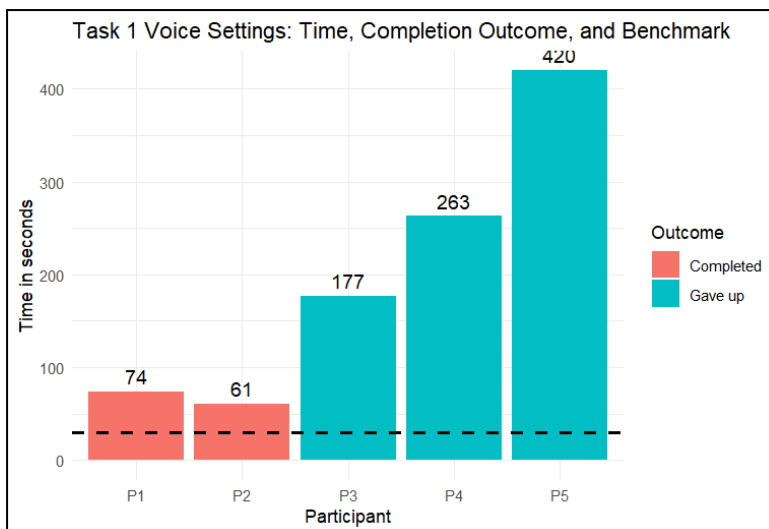


Figure 2.1.1.2 Task 1 time per participant compared with benchmark

Figure 2.1.1.2 compares individual completion times with the benchmark. All participants exceeded the benchmark time, including those who completed the task. The combination of low completion and long times indicates that the voice settings were not quickly understandable and that participants needed extended exploration before either succeeding or abandoning the task.

The quantitative results alone do not explain why participants struggled. Qualitative observations (Table 2.1.1) show that several participants initially treated the situation as a microphone or volume problem rather than a transmission-mode issue. They pressed the microphone mute button, adjusted input volume, or toggled unrelated options such as voice isolation instead of changing the voice activity mode. This behaviour suggests that the interface does not clearly convey the distinction between microphone availability and how speech is transmitted.

Table 2.1.1. Qualitative codes for Task 1 (voice settings)

Code name	Meaning	Example from data
Microphone/mode confusion	Participant treats the problem as a microphone or volume issue instead of a transmission-mode issue	P4 increases the microphone input volume and checks the voice input settings, but does not change the transmission mode.
Random setting exploration	Participant tries multiple unrelated options or panels without a clear strategy	P3 “presses some random buttons on the left-hand side” after failing to be heard in the call.
Push-to-talk not understood	Participant finds the push-to-talk option, but does not understand that a key must be held to speak	P2, P3, P4, P5 toggles push-to-talk but still cannot broadcast and continues changing other options such as deafen and stream attenuation.
Needs assistance	The participant only reaches a working configuration after a direct clue from the researcher	P2 can broadcast successfully only after being asked to check the settings button and being guided to push-to-talk.
Gives up	Participant explicitly stops trying, or the task is marked as abandoned	P3, P4 and P5 are marked as “gave up” after several unsuccessful attempts to fix the call.
Self-blame	Participants blame themselves or feel stupid rather than recognising the interface problem	P3 states that the test made them feel “very dumb” even though the issue was the push-to-talk configuration.

Participant 3’s session illustrates this pattern. They reached the voice settings page and even activated broadcasting by pressing the spacebar, but did not connect this action with the push-to-talk setting. Later, the same participant commented that the task made them feel “very dumb”. This combination of confusion and self-blame indicates that the system feedback around push-to-talk was not explicit enough for users to form a correct mental model of how voice transmission works.

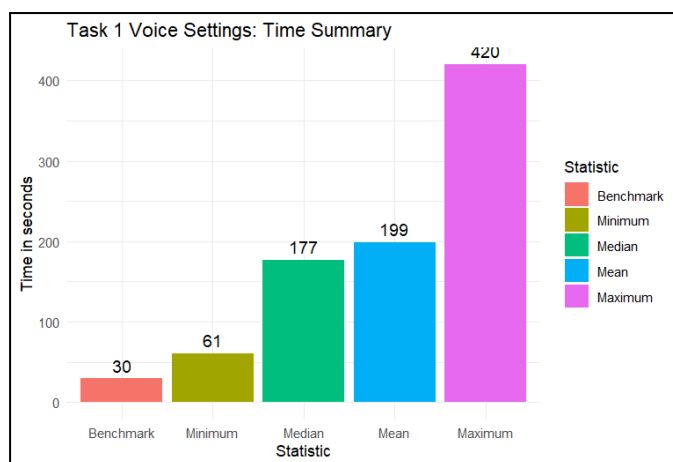


Figure 2.2.1.3. Task 1 time summary statistics

Figure 2.2.1.3 summarises Task 1 times using minimum, median, mean, and maximum values compared with the benchmark. The large gap between the benchmark and the median and mean times confirms that, on average, participants needed substantially more time than expected to configure voice transmission correctly.

2.1.2 Concept 2: Efficiency and Findability of Communication Hierarchy

Concept 2 was assessed across five communication sub-tasks. The analysis focused on whether participants could accurately distinguish between Discord's servers, channels, direct messages, mentions, and search results, especially when similar names created potential confusion.

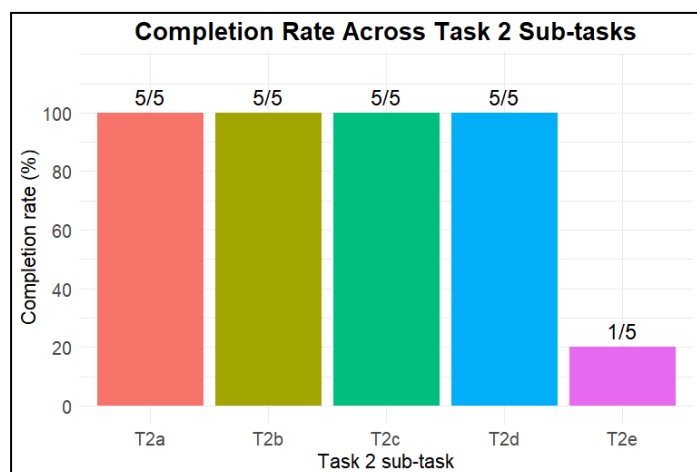


Figure 2.1.2. Completion rate across Task 2 sub-tasks

Figure 2.1.2 shows that the first four sub-tasks (T2a–T2d) were completed by all participants. Only the final sub-task, T2e, had a much lower completion rate, with one out of five participants successfully sending a message in the correct channel. This suggests that basic communication actions matched participants' expectations, while distinguishing between similarly named Discord's hierarchy did not.

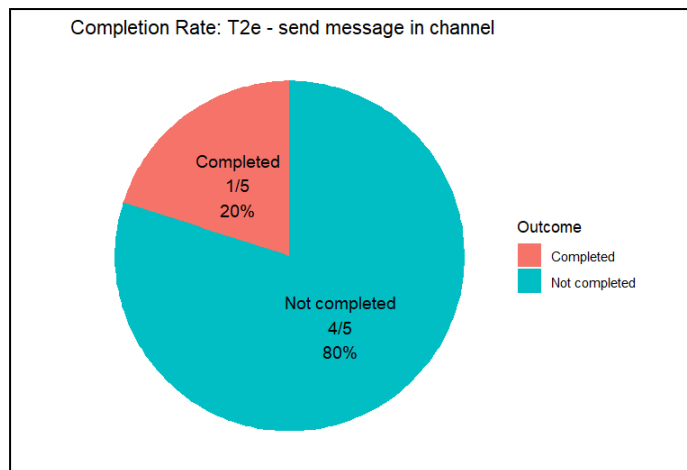


Figure 2.1.2.2 Completion task 2e - send message in channel “Usability testing”

Figure 2.1.2.2 focuses on T2e and confirms that four out of five participants did not complete the task. Most of these participants sent the message to the server rather than to the channel with the same name, highlighting confusion between levels of the hierarchy.

Table 2.1.2. Qualitative codes for Task 2e (Usability Testing channel)

Code name	Meaning	Example from data
False completion belief	Participant is convinced the task is done, while the message is actually in the wrong location	P2 says they “believe” they have already done it, although the message is in the server, not the channel.
Guessing location	Participant reports just guessing where to send the message rather than understanding the structure	P4 and P5 describe themselves as “just guessing” when trying to find the Usability Testing channel.
Relies on prior task	Participant assumes that being in the same place as the previous task means the new task is also correct	P4 stays in the server used for the previously mentioned task and does not initially notice that the next task requires a channel
Slow, hesitant navigation	Participant moves slowly and checks multiple servers or menus before finding (or failing to find) the right channel	P5 hovers between servers and channels, presses several icons on the left, and still sends the message in the server rather than the channel.

The qualitative data show that Participant 2 did not complete T2e because they believed the message in the server already satisfied the task, and Participant 4 assumed they were still in the correct place because the previous task also involved “Usability Testing”. These examples illustrate how overlapping names led participants to rely on recognition and memory of earlier steps instead of verifying the actual location in the hierarchy.

2.1.3 Concept 3: Findability of Server Management Features

Concept 3 examined whether participants could create a new server and then change their display name only inside that specific server. Quantitative measures included completion, mistakes, and time per sub-task; qualitative analysis focused on how participants interpreted server-level versus user-level settings.

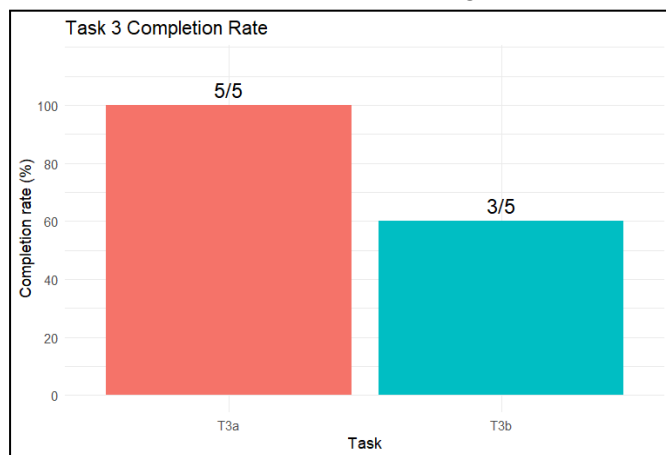


Figure 2.1.3. Task 3 completion rate

Figure 2.1.3 shows that all five participants completed Task 3a (creating a server), while only three completed Task 3b (changing the server-specific display name). This indicates an observable difference in difficulty between locating the Add Server button and finding the per-server profile option.

Changing the server-specific display name was more difficult. Participants often looked in server settings, members, roles, or account settings before finding the correct per-server profile option. This indicates that the problem was not creating a server, but locating the correct identity setting after the server had been created.

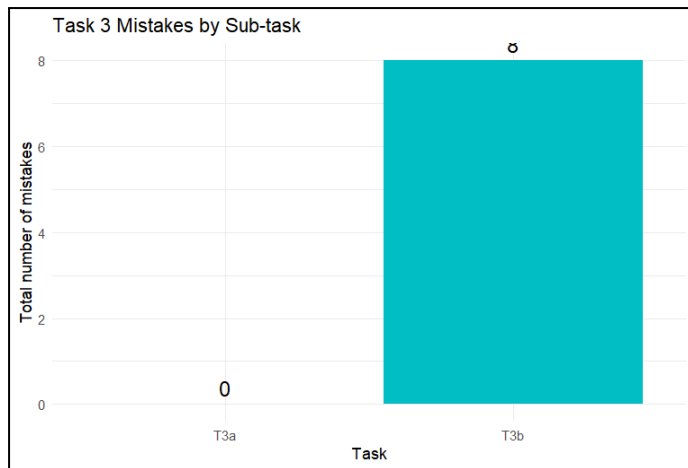


Figure 2.1.3.1. Mistakes across Task 3 sub-tasks

Figure 2.1.3.1 presents the number of mistakes for each sub-task. No mistakes were recorded for Task 3a, whereas eight mistakes occurred in Task 3b, reinforcing the conclusion that the identity-management step was much harder than server creation.

The qualitative data shows that some users were uncertain about the difference between global identity settings and server-specific identity settings.

Table 2.1.3. Qualitative codes for Task 3a (server creation)

Pattern / Code	Participants	Summary
Positive - Efficient Task Completion (Add Server button)	P1, P2, P3, P4, P5	All five found the Add/Create Server button quickly with no difficulty
Mental Model - Feature Purpose Unclear (server type options)	P1, P2, P3, P4, P5	All five participants were confused by the community vs friends vs study group options; no explanation was provided in the UI
Navigation - Avoidance Behaviour	P1, P3	Both chose to skip the server purpose question rather than decide
Mental Model - Inference Attempt	P3	Reasoned aloud about private vs public, tried to rationalise options without guidance

Task 3a shows that while the Add Server entry point itself was easy to find, the classification options within the creation flow were poorly understood. Participants advanced by skipping or guessing rather than by making informed choices.

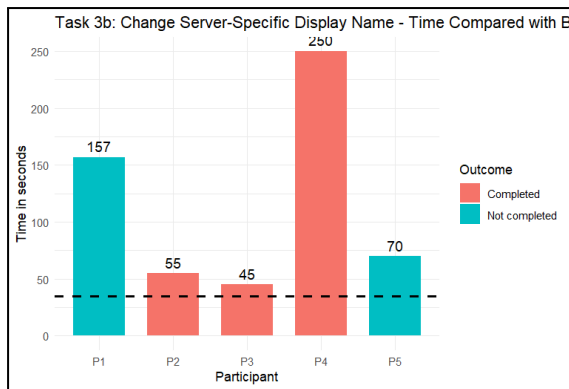


Figure 2.1.3.2. Task 3b time per participant compared with the benchmark

Figure 2.1.3.2 compares individual times for Task 3b with the benchmark. Participant 4 took substantially longer than the others because they narrated their reasoning in detail during the think-aloud, which inflated the recorded time. Across participants, Task 3b consistently took longer than expected, reflecting difficulty in locating the server-specific display-name option

Table 2.1.3.2 Qualitative codes for Task 3b (server-specific display name)

Code name	Meaning	Example from data
Wrong settings panel	Participant opens server-level settings when the task requires changing their own per-server profile	P1 repeatedly navigates to “Server settings” and edits the server name instead of their display name.
Server vs user name confusion	Participant confuses the server’s name with their own username or nickname	P1 changes the server title to “Work Project” and believes this satisfies the requirement to change their name.
Global vs per-server identity confusion	Participant is unsure whether changes apply only in one server or across all servers	P4 finds “edit per-server profile” but says they do not understand what “per server” means or which scope it affects.
Slow, uncertain exploration	Participant moves through multiple menus (members, roles, access, account settings) without a clear plan	P2 and P4 scroll through members, roles, engagement and other panels before eventually finding the correct profile option.
Verification behaviour	Participant actively checks other servers to confirm whether the change is local or global	P2 and P3 send messages or inspect other servers to verify that their main username still appears as “Test account”.
Persistent confusion after success	Participant completes the task but still reports feeling unsure or that they were “just guessing”	P4 states they are still confused and that they “were just guessing” even

		after successfully setting the server nickname.
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These codes show that the main problem in Task 3b was not clicking the wrong control by accident, but uncertainty about which layer of the system was being edited. Participants moved between server, account, and profile settings without a clear conceptual model of how identity changes propagate across servers.

2.2 Implications

To connect the qualitative analysis with the implications, the coded observations were summarised across participants. Figure 2.2 shows how often each qualitative code appeared in the sessions. This does not represent statistical generalisation, because the sample size was small, but it helps identify which usability problems appeared repeatedly across the tested users.

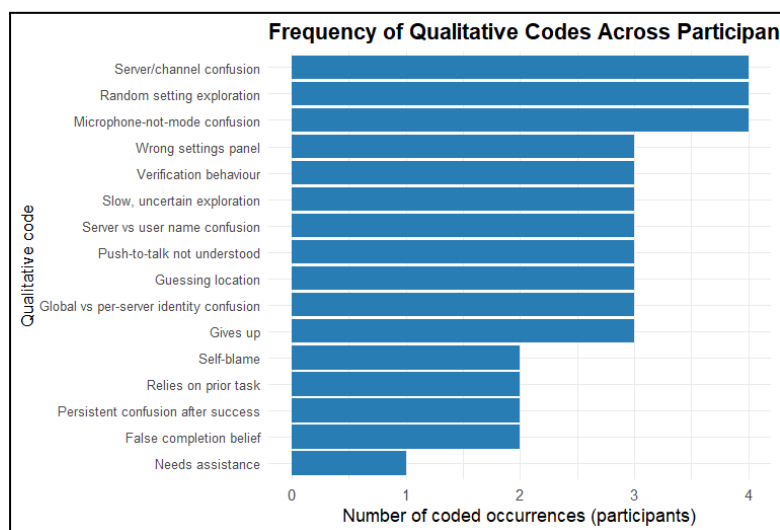


Figure 2.2: Frequency of qualitative codes across participants

For Concept 1, Understandability of Voice Communication Settings, the main implication is that Discord's voice settings need clearer feedback when push-to-talk or voice activity is active. This implication is supported by the low completion rate, long task times, and repeated mistakes in Task 1. It is also supported by Figure 2.2, where microphone/mode confusion, random setting exploration, push-to-talk not understood, and give-up behaviour appear across several participants. These patterns suggest that participants were not simply slow, but they misunderstood what the setting controlled. A future design improvement could show a visible message such as "Push-to-talk is active: hold [key] to speak". This would reduce random exploration of unrelated microphone settings and help users recover before giving up.

For Concept 2, Efficiency and Findability of Communication Hierarchy, the results suggest that familiar communication actions were usable, but Discord's server/channel hierarchy

became confusing when names overlapped. The strongest quantitative evidence is Task 2e, where only one participant sent the message in the correct channel. Figure 2.2 further supports this implication because server/channel confusion, false completion belief, reliance on the prior task, and slow hesitant navigation appeared in the qualitative coding. This suggests that participants recognised the name “Usability Testing” but did not always verify whether they were in the server or the channel. A future design improvement could make the current location more explicit.

For Concept 3, Findability of Server Management Features, the results show that creating a server was easy, but changing a server-specific display name was less findable. This is supported by the difference between Task 3a and Task 3b, where participants completed server creation more successfully than the server-specific display-name task. Figure 2.2 also supports this finding because wrong settings panel, server versus user name confusion, global versus per-server identity confusion, and slow, uncertain exploration appeared across multiple participants. This indicates that users were unsure which layer of the system they were editing: the server, their global profile, or their profile inside one server. A future design improvement could separate these actions more clearly. This would reduce the risk of users editing the wrong name or misunderstanding whether the change applies globally or only inside one server.

Overall, Figure 2.2 strengthens the interpretation that the main usability issue was not basic navigation alone. The repeated qualitative codes show that participants often formed the wrong mental model of Discord’s structure. Future iterations should therefore focus on clearer labels, stronger location indicators, and contextual feedback for settings that affect only one part of the system.

3 Limitations and Recommendations

3.1 Threats to Validity

Every design decision in a usability study introduces some risk to the reliability of the results. The following sections examine each key element of the study design and discuss the biases or limitations each may have introduced.

Tasks

The tasks were designed to test specific subsystems of Discord, but some were not detailed enough to fully cover the concept they aimed to measure. The main goal of Task 2 was to determine whether users could distinguish between a server and a channel. To avoid placing the two key steps back-to-back, which could signal to participants that they were being tested on exactly that distinction, several smaller tasks were inserted in between. While this was a reasonable design choice, it introduces its own threat to validity. The additional steps may have overloaded participants, leaving them less focused by the time they reached the critical navigation steps. A participant who struggled with Task 2 may have done so partly because of mental fatigue from the surrounding tasks rather than a genuine lack of understanding of Discord’s hierarchy. This makes it harder to isolate the specific cause of any errors observed during that task.

Task 3 had a similar issue. The steps covered server creation and identity management, but the connection between them was not always obvious to participants. Some may have completed individual steps without understanding why settings behave differently at the server level than at the account level. Breaking the task down further and asking participants to explain what they expected to happen at each step would have given more reliable data on their actual understanding.

Think-Aloud Protocol

The think-aloud protocol was used across all three tasks to capture verbal responses in real time. This method has two limitations. First, some participants may not have said everything that was going through their minds. The data collected is therefore incomplete in places, as silent confusion or hesitation is not always captured by what a participant says aloud. This means the qualitative data likely reflects only part of each participant's experience. Second, being asked to talk while completing a task can itself be a distraction. Participants who struggled with multitasking may have taken longer, not because the interface was confusing, but because their attention was split between doing the task and narrating it. This directly affects the time-on-task metric. One participant demonstrated this clearly during Task 3: the user stopped interacting with the interface entirely and began explaining her thoughts instead. The time recorded for her included this pause, inflating her result compared to others'. The data was kept as recorded rather than adjusted, since changing individual times would have made comparisons across participants unreliable. However, her result should be interpreted with this in mind.

Use of a Test Account on an Unfamiliar Device

All participants used a provided test account on a device that was not their own. For many participants, this was a source of unfamiliarity that went beyond Discord itself. Experienced Discord users tend to have their own settings, themes, keybinds, and layout preferences. Using an account with default settings on a different machine removes all of that. This is particularly relevant for the Discord platform, which is highly customisable. A participant who uses push-to-talk on their own account with a specific keybind may have been confused by the test account's default settings, not because the feature is hard to find, but because it looked different from what they were used to. This could have inflated error counts and task times, specifically in Task 1. A better approach would have been to allow participants to use their own accounts, though this would require different preparation to ensure consistent test conditions.

Rushing During the Session

Some participants appeared to rush through parts of the study, particularly during the post-task discussion. This may have been due to time pressure, discomfort with being observed, or simply wanting to finish. Participants who rushed through their verbal responses gave less detailed qualitative data, which makes it harder to understand the reasoning behind their actions.

No Familiarisation Period

Participants were given the test account and moved straight into the tasks without any time to explore the interface first. The account already had servers joined and a contact in the friends list, which were elements that participants had not set up themselves. In some cases, this may have caused brief confusion at the start of a task, as participants were navigating an account that looked partly set up but was not their own. Allowing a short familiarisation period before the tasks begin would give participants a chance to orient themselves, which would make the task results more representative of how they actually use the platform rather than how they cope with an unfamiliar account setup.

Sample Size

Only five participants took part in the study. This is a small sample and limits how much can be concluded from the quantitative data. Patterns observed across five participants may not hold across a wider population, and a single outlier has a proportionally large effect on averages. While five sessions can be enough to surface common usability issues, the data cannot be treated as statistically representative. A larger sample, ideally eight to ten participants at minimum, would produce more reliable results and make it easier to identify consistent patterns across different users.

Observer Presence and Recording

Screen recording was active throughout all sessions. For some participants, knowing they were being observed and timed introduced a degree of pressure that would not exist during normal Discord use. This could have led to rushed decisions, over-explaining, or more cautious behaviour than usual. This type of bias is difficult to remove entirely, but it does mean the results reflect how participants perform under observation rather than in a relaxed, everyday context.

User Profile

Only one user profile was used in this study, focusing on experienced Discord users. This was a deliberate decision to control for the variability that new users would introduce, but it limits how far the results can be applied. The findings reflect the experience of users who already know the platform to some degree. Usability issues that would only appear for a first-time user, such as not knowing what a server is at all, are not captured here. A future study including both profiles would give a more complete picture of where Discord's interface succeeds or fails.

Within the experienced user group, there was also variation in how participants interpreted the profile criteria. Some participants had used Discord mainly for gaming and had little experience with voice settings or server management in a work context. Others used it regularly for study or professional communication. This meant the baseline was not as consistent as intended, which may explain some of the differences in performance across participants.

3.2 Alternative Study Design

Based on the findings, a future version of the study should separate the three main concepts into shorter independent sessions rather than asking each participant to complete all tasks in one sequence. This would reduce fatigue, especially before the critical Task 2e server/channel distinction. The revised design would also include a short familiarisation period with the test account, followed by separate tasks for voice settings, communication hierarchy, and identity settings. For Task 3, the verification of whether the nickname change is local or global should be tested as a separate sub-task. This would make it easier to distinguish whether users can find the setting from whether they understand its scope.